

Adib Rouf

203-504-0685 | adibfrouf@gmail.com | linkedin.com/in/adibfrouf | github.com/AdibRouf

EDUCATION

University of Connecticut

Bachelor of Science in Computer Science
Concentration: Artificial Intelligence

Storrs, CT

Expected May 2027

Relevant Coursework: Data Structures, Algorithms & Complexity, Machine Learning, Artificial Intelligence, Systems Programming, Computer Architecture

EXPERIENCE

Data Engineering Intern

May 2026 – Aug. 2026

The Travelers Companies

Hartford, CT

- Modernized a legacy Active Directory workflow using Python, SQL, and parallel processing, resolving 200,000+ user-role records across 8,500+ groups and increasing processing speed by 500%.
- Automated approximately 80% of a recurring AutoSys audit workflow using Python and SQL, scanning approximately 800 jobs across 8 servers and tracing generic executing IDs to ownership and line-of-business data, reducing audit turnaround from hours or days to minutes.
- Prepared a 6-document internal knowledge corpus for a generative AI search prototype designed for an LLM, enabling context-aware retrieval beyond keyword matching.

Researcher

Jan. 2026 – Apr. 2026

NASA L'SPACE

Remote

- Applied requirements engineering and systems analysis throughout a 10-week proposal cycle to evaluate a spaceflight materials-testing protocol against mission-duration constraints and translate findings into testable engineering requirements.
- Led benchmark analysis comparing a bioactive dental-varnish concept with existing spaceflight dental care, evaluating effectiveness and retention to define technical requirements for the final NASA proposal.
- Collaborated on an 11-member multidisciplinary team of 9 undergraduate researchers, a NASA engineer, and a DDS to combine clinical benchmarks and mission constraints into the proposal's validation strategy.

Undergraduate Researcher

Mar. 2025 – Dec. 2025

University of Connecticut

Storrs, CT

- Engineered a full-stack ML pipeline using React, Flask, and Supabase that expanded the research platform from a fixed video library to user-uploaded video and saliency data, triggering HPC-based CLIP/Llama inference.
- Designed a 3-state asynchronous job workflow using Flask APIs and Supabase, synchronizing pending, processing, and complete states between the React frontend and HPC inference backend.
- Built the React results interface and Flask API integration to render source video, saliency heatmaps, and AI-generated explanations side by side, supporting end-to-end model evaluation and stakeholder demonstrations.

PROJECTS

ML Serving Platform

Jun. 2026 – Present

Python, FastAPI, Async Job Queues

- Built an ML inference API with FastAPI and asynchronous job queues, enabling non-blocking request handling through polling-based status tracking.
- Implemented dynamic request batching in the worker pipeline, increasing inference throughput by 88%, from 56 to 106 requests per second under concurrent load.
- Decoupled API request handling from model execution through a worker-based architecture, allowing the inference layer to scale independently.

Data Aggregation and Insights Platform

Dec. 2024 – Mar. 2025

Python, pandas, Snowflake

- Built a Python and pandas pipeline processing 28,900+ XML documents and approximately 40 GB of associated media, normalizing 100+ source columns into 15 standardized fields for downstream analysis.
- Automated ingestion of cleaned and normalized datasets into Snowflake, creating a repeatable workflow for storage, querying, and downstream analytics.
- Standardized episode-to-show relationships using GUID-based identifiers, linking each episode to a unique show ID and enabling reliable retrieval of all episodes associated with a given show.

TECHNICAL SKILLS

Languages: Python, SQL, C, JavaScript, HTML/CSS

Frameworks & Libraries: FastAPI, Flask, React, pandas, NumPy

Data, Cloud & Systems: Snowflake, Supabase, Azure, Git, Active Directory, AutoSys, HPC

ML & Backend: CLIP, Llama, LLM Inference, REST APIs, Async Job Queues, Parallel Processing